KOR STRUCTURAL · INTERNAL BRIEF

Six Years of Standards Chaos — Ended by Actually Standardizing

How one week produced a single ratified standard, a working virtual drafter, and a live standard-details library · Ian Lalonde · July 31, 2026

The problem, honestly stated

For at least six years KOR has run **two parallel standard-detail libraries** — CAD and Revit — with no declared master, no change control, and no one owning the whole. The Revit set drifted under years of unsupervised edits (wording rewritten, notes reshuffled, units added that engineering had explicitly rejected, four versions of the company logo in circulation). Three people independently tried to fix it — Simon's line-by-line comparison in May, Kevin's 153-comment engineering review in June, Rory's unit rulings — and the three reviews were never merged, because merging them was weeks of clerical labor nobody had. Everyone knew. Nobody could fix it alone. That is the definition of an organizational problem, not a drafting problem.

What actually fixed it: authority + machinery

The fix took two ingredients that had never been present at the same time:

1. An authority chain, in writing. Jim (principal): *Revit is the master; CAD is generated from it; only one version may exist.* Formatting delegated to Simon. Content questions answered by the engineers — Rory line-by-line with exact wordings, Kevin's review absorbed in full, Jim's approvals on the remaining items. Every open question in the whole mess now has a named ruling by a named person.

2. A machine to do the labor that burned the humans out. Every difference between the two libraries found mechanically (~2,000 divergent lines — the human reviews had reached perhaps a tenth of them); the three reviews merged; every line classified against the rulings; the residue routed back to humans as short, one-word decisions placed inside their own documents, in their own tools. Nobody re-reviewed anything; nobody parsed a spreadsheet; the people supplied only judgment.

> **The one rule that made the politics work:** each person's existing review was accepted in full — no one was ever asked to re-approve their own work. Questions went to people as one-word decisions at the exact spot in their own marked-up PDF. Engineers engineer, drafters draft, the machine does everything else.

Deliverable 1 — The Standard, actually standardized

- **One master** (Revit), being produced now in a single automated batch: cleaned of every engineer-rejected addition (all kN references, the killed kPa presentations, the odd metric steel/weld sizes), corrected at every engineer-confirmed error (crane-opening reinforcing, missing site-wall dimension, tie-wall value), formatted per the ratified calls (bolding, brackets, plain-digit metric, ordering, one logo).
- **CAD generated from the master** — drift is no longer policed; it is structurally impossible.
- **A standing automated re-comparison** re-checks the libraries so divergence can never silently return.
- **Versioned, with an assumptions sheet**: every provisional call is documented; the engineering committee's follow-up review lands as an addendum, not a restart.

Deliverable 2 — The Virtual Drafter, proven on real work

- Took a real engineer's marked-up PDF and produced the same model changes as the human drafter — and at the two locations where they differed, **the engineer confirmed the machine's version**. One of those was a wrong bar size already printing on issued sheets; it has since been corrected in production.
- Read the **entire model fleet** (198 models, ten Revit versions) overnight, read-only, building the per-job knowledge that makes every future task start informed — and surfacing fleet-wide defects (≈4,000 bar tags printing "NA" — root-caused, fix engineer-approved and in test).

- **Delivery pipeline built**: drop a marked-up PDF in a folder → the full job runs unattended in a sandbox → a branded package returns (change plan, executed changes, photographic before/after verification, checks) with zero human touches. Every task it runs banks evidence for the management demonstration.
- **Safety by construction**: it physically cannot write to a live project model; every change is verified by independent re-read; real errors stop and wait for a human. Production remains human-only until management has seen flawless runs — each shadow task adds to that portfolio.

Deliverable 3 — The Standard Details library, as living data

The ratified standard is not a PDF in a folder — it exists as structured, versioned content: every sheet, note, and detail with its ruling, its author, and its history. That is exactly the data a standard-details module needs: browsable by drafters, enforceable by the Virtual Drafter (its output is checked against the same source), and maintainable through one controlled channel instead of folklore. The library and the drafter share one source of truth — which is what keeps the standard standard.

By the numbers

198
models read fleet-wide, zero modified

~2,000
divergent standard lines found & classified

3
independent human reviews merged

~30
one-word human decisions — total labor asked of people

1
wrong bar size caught in issued work, engineer-confirmed

4,000
bad tag reads root-caused, fix approved

8/8
checks passed in the end-to-end rehearsal

0
writes ever made to a production model

What remains

Item	State
Execute the reconciliation batch (cleaning + corrections + formatting) on the Revit masters — sandbox	Staged; runs when the current test jobs complete
Engineering committee follow-up (Rory, next week)	Lands as addendum to the versioned standard
Regenerate CAD set from the reconciled master	After the batch; Simon operationalizes
Issuance wrapper (plotted sheet sets, revision clouds) + from-scratch drawing capability	Next build phase; Jim's upcoming drawing is the milestone
Management demonstration & production go-decision	When the proof portfolio says so — it grows with every shadow task

The bottom line: the mess persisted for six years because fixing it required simultaneously an authority to rule, and thousands of hours of labor to execute the rulings. The authority now exists in writing, and the labor is now machinery. That is why this time it is actually fixed — and why it stays fixed.